

## CONTACT INFORMATION

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## EDUCATION

**University of Sussex** 2023-

- PhD in Informatics  
Thesis: Considering spatial embeddings and synaptic delays for neural networks in neuromorphic computing

**ELTE Eötvös Loránd University** 2020-2023

- M.Sc. in Artificial Intelligence  
Thesis: Extending Sparse Dictionary Learning Methods for Adversarial Robustness  
Final Grade: *Excellent*

**Humboldt University Berlin** 2022

- Exchange semester with ERASMUS

**ELTE Eötvös Loránd University** 2016-2020

- B.Sc. in Computer Science, specialization in modelling  
Thesis: Generating Symbolic Music using Artificial Neural Networks  
Final Grade: *Good*

**Johannes Kepler University Linz** 2019

- Exchange semester with CEEPUS

## EXPERIENCE

### Research

- **Sussex AI** 2023-
  - Delay learning and structural plasticity
- **MARCS Institute** 2026
  - Visiting researcher at the ICNS
- **Cold Spring Harbor Laboratory** 2026
  - NeuroAI research intern
- **Centrum Wiskunde & Informatica** 2025
  - Visiting researcher with Sander Bohte
- **The Alan Turing Institute** 2024-2025
  - NeuroAI and backdoor reinforcement learning
- **Neural Information Processing Group** 2021-2022
  - Sparse approximation for adversarial robustness
  - Multimodal emotion estimation using transformer architectures

### Teaching

- **Neuromatch Academy** 2025
  - NeuroAI
- **University of Sussex** 2024
  - Bio-inspired AI
- **ELTE Eötvös Loránd University** 2020
  - Algorithms and Data Structures
  - Object-oriented programming in C++

### Software development

- **evosoft** 2019-2020
  - C++ Software Engineer

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|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| SKILLS                 | <ul style="list-style-type: none"> <li>• <i>Programming Languages</i>: C/C++, Python, Prolog</li> <li>• <i>Softwares/Libraries</i>: NumPy, Keras, PyTorch, OpenCV, Microsoft Office, Latex</li> <li>• <i>Language Skills</i>: English: C1 (Advanced), German: A2 (Elementary)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                  |
| AWARDS AND FELLOWSHIPS | <ul style="list-style-type: none"> <li>• MARCS International Visiting Scholar Fellowship</li> <li>• The Alan Turing Institute - PhD enrichment scheme</li> <li>• be.AI Leverhulme Doctoral Scholarship</li> <li>• Nokia Young Scientist Award</li> <li>• TDK - 1st prize</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026<br>2024<br>2023<br>2023<br>2022                                                             |
| CERTIFICATES           | <ul style="list-style-type: none"> <li>• Associate Fellow in Advanced HE</li> <li>• Neuromatch Computational Neuroscience Summer School</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2024<br>2023                                                                                     |
| PUBLICATIONS           | <ul style="list-style-type: none"> <li>• Journal <ul style="list-style-type: none"> <li>◦ Mészáros, B., Knight, J., &amp; Nowotny, T. <i>Nature Communications</i>. Efficient Event-based Delay Learning in Spiking Neural Networks</li> <li>◦ Ghosh, M., Habashy, K. G., De Santis, F., Fiers, T., Ercelik, D. F., Mészáros, B., ... &amp; Goodman, D. F. <i>eNeuro</i>. Spiking neural network models of interaural time difference extraction via a massively collaborative process</li> <li>◦ Mészáros, B., Knight, J., &amp; Nowotny, T. <i>Frontiers in Computational Neuroscience</i>. Learning Delays Through Gradients and Structure: Emergence of Spatiotemporal Patterns in Spiking Neural Networks.</li> </ul> </li> <li>• Conference <ul style="list-style-type: none"> <li>◦ Mészáros, B., Knight, J. C., Timcheck, J., &amp; Nowotny, T. <i>International Conference on Neuromorphic Systems (ICONS)</i>. A Complete Pipeline for deploying SNNs with Synaptic Delays on Loihi 2</li> <li>◦ Shoesmith, T., Knight, J. C., Mészáros, B., Timcheck, J., &amp; Nowotny, T. <i>Neuro Inspired Computational Elements (NICE)</i>. Eventprop training for efficient neuromorphic applications</li> <li>◦ Szeghy, D., Aslan, M., Fóthi, Á., Mészáros, B., Milacski, Z. Á., &amp; Lőrincz, A. <i>International Conference on Deep Learning Theory and Applications (DeLTA)</i>. Structural Extensions of Basis Pursuit: Guarantees on Adversarial Robustness.</li> </ul> </li> <li>• Submitted <ul style="list-style-type: none"> <li>◦ Mészáros, B., Knight, J. C., Akarca, D., &amp; Nowotny, T. Space as Time Through Neuron Position Learning. Submitted to <i>Nature Machine Intelligence</i>.</li> </ul> </li> <li>• In preparation <ul style="list-style-type: none"> <li>◦ Pehle, C., Mészáros, B., &amp; Zador, A. In preparation for <i>Nature</i>.</li> </ul> </li> </ul> |                                                                                                  |
| ACADEMIC OUTREACH      | <ul style="list-style-type: none"> <li>• Talks <ul style="list-style-type: none"> <li>◦ Budapest Computational Neuroscience Forum</li> <li>◦ Spiking Neural Networks as Universal Function Approximators</li> <li>◦ International Workshop on Reliable and Sustainable Neuromorphic Hardware</li> <li>◦ Neuro Inspired Computational Elements Conference</li> <li>◦ Sensory Neuroscience and Computation centre seminar</li> <li>◦ Robots in Disguise: Fundamental AI Reading Group</li> <li>◦ MPIE Conference</li> <li>◦ OTDK - National Scientific Student Body Conference</li> </ul> </li> <li>• Posters <ul style="list-style-type: none"> <li>◦ Spiking Neural Networks as Universal Function Approximators</li> <li>◦ 33rd annual Computational Neuroscience Meeting</li> <li>◦ The Fifth International Convention on the Mathematics Of Neuroscience and AI</li> <li>◦ International Conference on Neuromorphic Computing and Engineering</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2026<br>2025<br>2025<br>2025<br>2025<br>2024<br>2024<br>2023<br><br>2024<br>2024<br>2024<br>2024 |

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| ADDITIONAL | • Welfare at <b>FOLD</b>                                                     | 2025-     |
|            | • Volunteer work                                                             |           |
|            | ◦ <b>A Város Mindenkié</b>                                                   |           |
|            | ◦ <b>Heti Betevő</b>                                                         |           |
|            | ◦ Organizing AI themed artisan workshops for layman                          |           |
|            | • Sport activity                                                             |           |
|            | ◦ Sabre fencing - 4 times national champion, European Championship 4th place | 2005-2016 |
|            | • Music production, Songwriting, DJing                                       |           |
|            | ◦ Debut EP forthcoming on <b>SCORDISCI</b>                                   | 2026      |